Factors dominating adhesion of NaCl onto potato chips.

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In this study, the adhesion factors examined were time between frying and coating, surface oil content, chip temperature, oil composition, NaCl size, NaCl shape, and electrostatic coating. Three different surface oil content potato chips, high, low, and no, were produced. Oils used were soybean, olive, corn, peanut, and coconut. After frying, chips were coated immediately, after 1 d, and after 1 mo. NaCl crystals of 5 different particle sizes (24.7, 123, 259, 291, and 388 microm) were coated both electrostatically and nonelectrostatically. Adhesion of cubic, dendritic, and flake crystals was examined. Chips were coated at different temperatures. Chips with high surface oil had the highest adhesion of salt, making surface oil content the most important factor. Decreasing chip temperature decreased surface oil and adhesion. Increasing time between frying and coating reduced adhesion for low surface oil chips, but did not affect high and no surface oil chips. Changing oil composition did not affect adhesion. Increasing salt size decreased adhesion. Salt size had a greater effect on chips with lower surface oil content. When there were significant differences, cubic crystals gave the best adhesion followed by flake crystals then dendritic crystals. For high and low surface oil chips, electrostatic coating did not change adhesion of small size crystals but decreased adhesion of large salts. For no surface oil content chips, electrostatic coating improved adhesion for small salt sizes but did not affect adhesion of large crystals.